

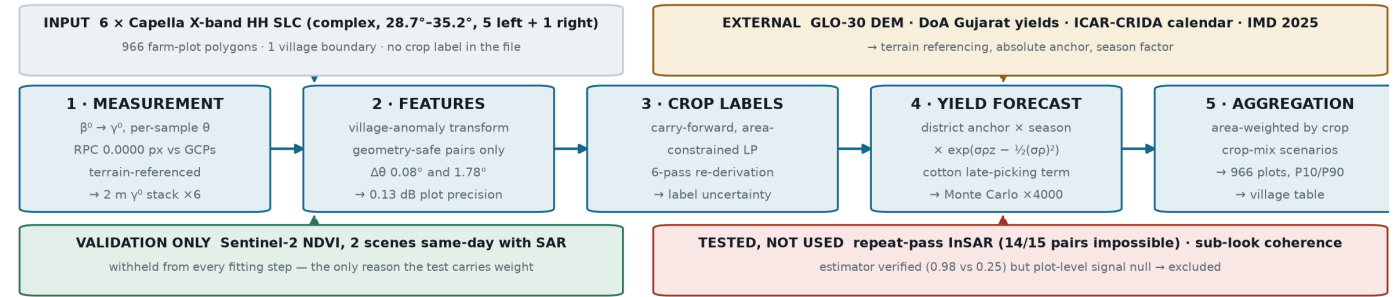
Six Looks, One Harvest

A final at-harvest yield forecast for 966 farms in Sokhda, Vadodara — built on what X-band can actually measure, and explicit about what it cannot.

Project Title: Six Looks, One Harvest · **Team Name:** Team 8bit — Harsh Thummar, Viraj Suhagiya · **Primary Domain:** Remote Sensing – Precision Agriculture / Crop Yield Forecasting

Data Sources Used: Capella X-band HH stripmap SLC ×6 (6/19 Jun, 14 Aug, 13/29 Oct, 12 Nov 2025) · farm & village shapefiles · Copernicus DEM GLO-30 · Directorate of Agriculture Gujarat district yields · ICAR-CRIDA Vadodara Contingency Plan · IMD/Gujarat SEOC 2025 monsoon · Sentinel-2 L2A (**validation only, never fitted**)

PLOTS FORECAST 966/966 832 with all six dates	RPC GEOCODING 0.0000 px vs the 225 embedded GCPs	TERRAIN ROUND-TRIP 4.1–4.7 px was 15–17 px flat-earth	VS SAME-DAY NDVI ρ +0.56 n = 923, 13 Oct	STANDING CROPS ONLY ρ +0.36 rice & cotton, p < 1e-8
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Page 1: Problem, Strategy, and Novelty

1 Proposed Strategy & Technical Novelty

AI methodology. This track publishes **no plot-level ground-truth yield** — not withheld, absent. A supervised regressor has nothing to fit: an off-the-shelf regression baseline is not weaker here, it is *undefined*. Round 2 answered that with two *relative* indices; Round 3 must return *absolute* kg/ha at harvest. So radar sets structure, official statistics set level:

$$Y_p = Y_{\text{ref}}(c) S(c) \exp(\sigma_c \rho z_p - \frac{1}{2}(\sigma_c \rho)^2)$$

z_p is a stage-weighted SAR index (establishment · 13 Oct canopy · 12 Nov retention · evenness); σ_c the farm-to-farm yield CV inside a village; ρ the share of it six X-band passes explain. Crop assignment stays an **area-constrained transportation LP** against phenological templates — not a trained classifier.

Novel contribution — the twist.

- **The lognormal form is the agronomic plausibility constraint.** It forces the *area-weighted* village mean to reproduce $Y_{\text{ref}} \cdot S$ **by construction**, so the level is a declared assumption and everything beneath it is SAR-driven. The model is structurally unable to invent a village yield the radar cannot support.
- **A second geometry-safe pair, new to Round 3.** The added passes create 13 Oct→12 Nov ($\Delta\theta$ 1.78°, same look) — the axis separating a crop still standing from one already cut. Round 2 had only 19 Jun→14 Aug (0.08°).
- **The right-looking pass as a terrain diagnostic.** A height error displaces opposite look directions in *opposite* senses, so a residual on 29 Oct alone is a terrain signature, not a registration failure — and it caught our own regression (§4).
- **An explicit forecast term.** Cotton alone still stands at the last pass; its final lint carries a late-picking multiplier from that plot's own 13 Oct→12 Nov retention (0.79–1.27).

Success metrics — proving it without labels.

- **A falsifiable physical prediction:** the index must correlate where a crop still stands and go *null* where it has been cut. An artefact correlates everywhere or nowhere (§3).
- **Independent same-day optical:** two Sentinel-2 scenes are same-day coincident with Capella passes, and NDVI enters no fitting step — the only thing making the test meaningful.

- **Geometric self-consistency** (RPC error vs the 225 GCPs, co-registration residual, agreement with Capella's GEO product) and **interval calibration** — village interval width must order with how much of each crop's cycle was observed.

2 Ancillary Grounding & Assumptions

- **Stated assumption.** SAR sets *ranking and spread*; district yield × season factor sets the *level*. Sokhda is assumed not to depart from its district mean, as nothing in the data justifies claiming otherwise. $\rho = 0.55$ (varied 0.40–0.70) is assumed, not estimated — the least-constrained parameter in the model.
- **Datum, carried forward and extended.** Capella RPCs expect *ellipsoidal* heights; GLO-30 stores *orthometric*. The offset is *solved* from the 225 GCPs per product, now across six scenes: $N = -62.02$ m, scene-to-scene sd **0.12 m**, agreeing with the four-scene Round-2 solution to 0.06 m.
- **Known issue — swath coverage.** The AOI is wider than one Capella stripmap: 832 plots carry all six dates, 91 partial, **43 none**. The 43 get the crop mean with inflated uncertainty and a `not_observed_swath_edge` flag — shipped, not dropped.
- **Known issue — contaminated baseline.** Sentinel-2 shows **326 plots already green on 10 June**, so a naive per-plot “bare soil” reference is itself contaminated. Features are village-anomaly referenced instead.
- **Known issue — label provenance.** Crop labels are the Round-1/2 carry-forward: self-derived, not organiser ground truth, median confidence 0.16. Propagated as uncertain, not trusted (§4).

3 Preliminary Salient Results

Validation note. This track publishes **no plot-level ground-truth yield at any stage**, so RMSE / MAE / R^2 and a spatially held-out split are *not computable* — reporting them would require inventing labels. We report instead (a) agreement with an **independent sensor never used in fitting** and (b) internal geometric consistency. All 966 plots carry a forecast.

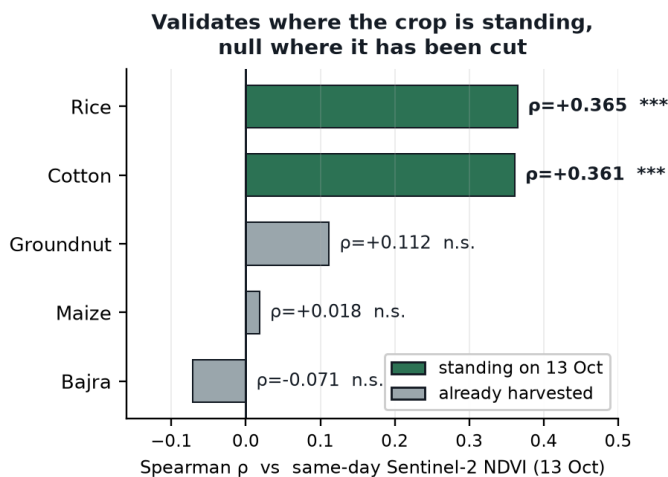
THE DISCRIMINATING RESULT

SAR yield index vs. *same-day* Sentinel-2 NDVI (13 Oct), computed *within* crop:

CROP	RHO	P	ON 13 OCT
Rice	+0.365	2e-08	standing
Cotton	+0.361	1e-12	standing
Groundnut	+0.112	0.079	harvesting
Maize	+0.019	0.88	harvested
Bajra	-0.071	0.61	harvested

Significant for the two crops still in the field; **statistically null for the three already cut**, where both sensors see bare soil. This is the \$1 prediction passing. Whole village, same day: $\rho = +0.56$ (13 Oct, $n=923$) and $+0.48$ (12 Nov, $n=937$).

Geometric progress, quantified. RPC forward model vs. the 225 embedded GCPs: **0.0000 px**. Restoring terrain referencing improved the GCP round trip from **15.1–16.6 px to 4.1–4.7 px**, raised correlation with Capella's own GEO product **0.769 → 0.840** on *6 of 6* passes, and collapsed the right-look co-registration residual **2.54 m → 0.12 m**.



Village-level forecast. Area-weighted, all 966 plots. Interval width orders correctly with cycle coverage — bajra widest at $\pm 20\%$, rice tightest.

CROP	HA	KG/HA	P10-P90	TONNES
Rice	60.4	1717	1401-2021	103.7
Cotton	192.7	820	674- 971	158.0
Maize	34.3	2479	2009-2968	85.0
Bajra	34.6	2932	2378-3576	101.4
Groundnut	125.6	2453	2005-2922	308.1
VILLAGE	447.5	--	--	756.1

4 Technical Challenges & Pivots

- No ground truth (structural).** Pivot: abandoned supervised regression outright for the anchored hybrid, and rebuilt validation around a withheld independent sensor.
- A 60-day gap, 14 Aug → 13 Oct**, brackets the *entire* maturity and harvest of maize, bajra and groundnut — sampled once. Not fixable; carried as wider intervals, not hidden.
- Pivot — we corrected a regression in our own pipeline.** The Round-3 chain was first rebuilt on a single constant height. All six passes agreed on it, residual ≤ 2.8 m, so it looked right. It was not: a global shift metric recovers the *mean* alignment and hides spatially varying terrain, and this block carries 26.6 m

of relief = up to 46 m of ground range. The 29 Oct right-look residual exposed it; terrain referencing — Round 2's standard — was restored and extended to six passes.

- Pivot — we rejected our own new classifier.** An independent six-pass crop map agreed with the carry-forward on only **46%** of observed plots. On withheld Sentinel-2 the *carry-forward separated better* (η^2 0.138 vs 0.085), so we kept it and converted the disagreement into measured label uncertainty.
- Extended, still null — the phase.** Round 2 ruled out repeat-pass InSAR on four scenes; with six, all **15 pairs** fail — 5 opposite-look (the new 29 Oct), 9 beyond the critical baseline, and the one viable pair ($B_{\perp}$ 844 m) 56 days apart. We re-ran sub-look coherence in the *azimuth* aperture rather than Round 2's range sub-bands: the estimator verifies (points $\gamma=0.98$ vs 0.25 clutter) and is again null at plot level ($r=0.02$, $\eta^2=0.02$). Null in both dimensions.

THE LARGEST OPEN RISK IS NOT THE RADAR

The Round-1 carry-forward puts **groundnut at 28% of Sokhda**; the Directorate of Agriculture puts it at **0.35% of the district**. Both cannot be close to right, and groundnut carries a high per-hectare reference. Running the assignment under both area constraints moves village production **763 t → 570 t (−25%)**, essentially all of it groundnut. We publish both scenarios rather than choose silently.

5 Final Sprint Roadmap

Immediate (next 10 hours). Upload the executed notebook to Kaggle and set it **public**; attach it plus the nine media-gallery figures, the written methodology and the Goa deck; select the Track and **hit Submit** — a saved draft is not judged. Final read-through of the 2,000-word writeup against the rubric.

Refinement (running / queued). Sensitivity of the village total to ρ across 0.40–0.70; re-running the crop-mix scenarios with the district constraint as primary, to bound the groundnut risk from both sides; rehearsing the 16-slide deck to the 10-minute limit.

Final output. A public end-to-end notebook (raw complex SLC → both output tables, verified to reproduce them *exactly* from a fresh kernel, closing with assertions); a plot-level CSV of 966 forecasts with P10/P90 and quality flags; a village table by crop; a machine-readable validation report; and the Goa presentation. **No trained model artefact — by design, because there is nothing to train on.**

Appendix. GitHub: <https://github.com/harshtummar77/AISEHack-2.0-ROUND-3>

Kaggle notebook: public at submission, linked in the Writeup's Project Files. **Data:** Capella X-band HH SLC $\times 6$ (competition) · Copernicus DEM GLO-30 tile N22E073, AWS Open Data · Sentinel-2 L2A via Element84 Earth Search, *validation only* · Directorate of Agriculture Gujarat, *Area, Production and Yield* (2024) · ICAR-CRIDA/AAU-Anand, *Agriculture Contingency Plan: Vadodara* · IMD/Gujarat SEOC 2025 monsoon summary. **References:** Attema & Ulaby (1978), *Vegetation modeled as a water cloud*, Radio Sci. 13(2) 357–364 · Parmar & Bhatt (2025), *Agricultural patterns and crop shifts in Vadodara-Chhotaudepur*, IJAFS 7(5) 55–61 · Monteith (1977), radiation-use efficiency · Capella Space, *SAR Imagery Products Format Specification* (radiometry, RPC, NESZ).